ELLIOT G MITCHELL

New York, NY
646-932-0430 ⋄ emitchell3@geisinger.edu

EDUCATION

Columbia University, New York, NY 2016 - 2021
Doctor of Philosophy (PhD), Biomedical Informatics
Dissertation: *Enabling Automated, Conversational Health Coaching with Human-Centered AI*

Carleton College, Northfield MN 2009 - 2013
Bachelor of Arts (BA), Psychology (with Distinction)
Concentration in Cognitive Science

WORK EXPERIENCE

Geisinger, Danville, PA (Remote from New York, NY)
Lead AI Product Manager 2026 - Present
Data Scientist Team Lead 2023 - 2026
Senior Data Scientist 2021 - 2023

Columbia University, New York, NY
Graduate Research Assistant 2016 - 2021

NourishedRx (fka Project Well), Stamford, CT (Remote)
Data Science and Product Management Consultant 2020 - 2021

Columbia Technology Ventures, New York, NY
Technology Ventures Fellow 2019 - 2019

Epic, Madison, WI
Technical Coordinator 2015 - 2016
Technical Services 2013 - 2016

Bon Appetit at Carleton College, Northfield, MN 2010 - 2013
Student Manager, Food Services

HONORS and AWARDS

Second Place NIH Long COVID Computational Challenge (L3C) 2023

Honorable Mention (2nd place) AMIA Doctoral Dissertation Award 2022

Top 3 Finalist Hearst Health Prize 2022

Stage 1 Finalist CMS AI Health Outcomes Challenge 2020

First Place AMIA Student Design Challenge 2017, 2018

Third Place HITLAB World Cup 2017

Top 5 Finalist Alexa Diabetes Challenge 2017

Phi Beta Kappa Carleton College 2013

Magna Cum Laude Carleton College 2013

Sigma Xi Carleton College 2013

Dean's List Carleton College 2011–2013

PUBLICATIONS

JOURNAL ARTICLES and CONFERENCE PROCEEDINGS

Elliot G Mitchell, Pooja Desai, Arlene Smaldone, Andrea Cassells, Jonathan N. Tobin, David Albers, Matthew Levine, and Lena Mamykina. 2025. T2 Coach: A Qualitative Study of an Automated Health Coach for Diabetes Self-Management. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. doi:10.1145/3706598.3714404

Pretko AD, Ryer EJ, Elmore JR, Salzler GG, Unrue E, Lewis AJ, Vawdrey DK, Maff RM, **Mitchell EG**. Clinical and Financial Impact of a Machine Learning Powered Screening Program for Abdominal Aortic Aneurysms. *JVS-Vascular Insights*. 2024 Oct 22. doi:10.1016/j.jvsvi.2024.100150

Ashika Farzana, Satish Kalepalli, Grant DeLong, Vishal Mehra, Emily Fry, David K Vawdrey, and **Elliot G Mitchell**. Integrating Remote Patient Monitoring Data into Machine Learning Models for Predicting Emergency Department Utilization. *American Medical Informatics Association (AMIA) Annual Symposium Proceedings*. November 2024. PMID: 40417519; PMCID: PMC12099409.

Bergquist T, Loomba J, Pfaff E, Xia F, Zhao Z, Zhu Y, **Mitchell E**, Bhattacharya B, Shetty G, Munia T, Delong G, et al. Crowd-sourced machine learning prediction of long COVID using data from the National COVID Cohort Collaborative. *EBioMedicine*. 2024 Oct 1;108. doi:10.1016/j.ebiom.2024.105333

Salzler GG, Ryer EJ, Abdu RW, Lanyado A, Sagiv T, Choman EN, Tariq AA, Urick J, **Mitchell EG**, Maff RM, DeLong G, Shriner SL, Elmore JR. Development and validation of a machine-learning prediction model to improve abdominal aortic aneurysm screening. *Journal of Vascular Surgery*. 2024 Jan 17. doi:10.1016/j.jvs.2023.12.009

Burgermaster M, Desai PM, Heitkemper EM, Juul F, **Mitchell EG**, Turchioe M, Albers DJ, Levine ME, Larson D, Mamykina L. Who Needs What (Features) When? Personalizing Engagement with Data-Driven Self-Management to Improve Health Equity. *Journal of Biomedical Informatics*. 2023 Jun 8:104419. doi:10.1016/j.jbi.2023.104419

Bhattacharya B, DeLong G, **Mitchell EG**, Munia TT, Shetty G, Tariq A. A Long COVID Risk Predictor Focused on Clinical Workflow Integration. *medRxiv*. 2023:2023-05. doi:10.1101/2023.05.26.23290243

Kocaballi AB, Sezgin E, Clark L, Carroll JM, Huang Y, Huh-Yoo J, Kim J, Kocielnik R, Lee YC, Mamykina L, **Mitchell EG**, Moore RJ, Murali P, Mynatt ED, Park SY, Pasta A, Richards D, Silva LM, Smriti D, Spillane B, Zhang Z, Zubatiy T. Design and Evaluation Challenges of Conversational Agents in Health Care and Well-being: Selective Review Study. *Journal of Medical Internet Research*. September 2022. doi:10.2196/38525

Mitchell EG, Elhadad N, Mamykina L. Examining AI Methods for Micro-Coaching Dialogs. *CHI 2022: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. April 2022. doi:10.1145/3491102.3501886

Liu LG, Grossman RH, **Mitchell EG**, Weng C, Natarajan K, Hripcsak G, Vawdrey DK. A deep database of medical abbreviations and acronyms for natural language processing. *Scientific Data*. 2021 Jun 2;8(1):1-9. doi:10.1038/s41597-021-00929-4

Mitchell EG, Maimone R, Tobin, JN, Cassells A, Davidson P, Smaldone AM, Mamykina L. Automated vs. Human Health Coaching: Exploring participant and practitioner experiences. *Proceedings of the ACM on Human-Computer Interaction. CSCW 2021* doi:10.1145/3449173

Mitchell EG, Heitkemper E, Burgermaster M, Levine M, Miao Y, Hwang M, Desai P, Cassells A, Tobin J, Tabak EG, Smaldone AM, Albers DJ, and Mamykina L. From Reflection to Action: Combining Machine Learning with Expert Knowledge for Nutrition Goal Recommendations. *CHI 2021: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* doi:10.1145/3411764.3445555

Mitchell EG, Tabak E, Levine M, Mamykina L, Albers D. Enabling Personalized Decision Support with Patient-Generated Data and Attributable Components. *Journal of biomedical informatics.* (2021) 103639. doi:10.1016/j.jbi.2020.103639

Turchioe MR, Burgermaster M, **Mitchell EG**, Desai PM, Mamykina L. Adapting the stage-based model of personal informatics for low-resource communities in the context of type 2 diabetes. *Journal of biomedical informatics.* 2020 Sep 20:103572. doi:10.1016/j.jbi.2020.103572

Desai PM, **Mitchell EG**, Hwang ML, Levine ME, Albers DJ, Mamykina L. Personal Health Oracle: Explorations of Personalized Predictions in Diabetes Self-Management. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems.* doi:10.1145/3290605.3300600

Mitchell EG*, Grossman LV*, Hripcsak G, Weng C, Vawdrey DK. A method for harmonization of clinical abbreviation and acronym sense inventories. *Journal of biomedical informatics.* 2018 Dec 1;88:62-9. doi:10.1016/j.jbi.2018.11.004

Grossman LV, Feiner SK, **Mitchell EG**, Creber RM. Leveraging Patient-Reported Outcomes Using Data Visualization. *Applied clinical informatics.* 2018 Jul;9(03):565-75. doi:10.1055/s-0038-1667041

BOOK CHAPTERS

Lena Mamykina, **Elliot G Mitchell**, Pooja Desai, and David Albers. Intelligent Decision Support in Personal Health: Personalized Health Coaching in Type 2 Diabetes. Published in *Human Computer Interaction in Healthcare: The Role of Cognition. Second Edition.* pp. 413-438. Eds. Andre W. Kushniruk, David R. Kaufman, Thomas G. Kannampallil, and Vimla L. Patel. Springer Cham. November 2024, doi:10.1007/978-3-031-69947-4_17

Elliot G Mitchell and Lena Mamykina. Artificial Intelligence to Support Self-Management and Coaching. Published in *Diabetes Digital Health, Telehealth, and Artificial Intelligence*, pp. 275-286. Eds. David C Klonoff, David Kerr, Juan Esponoza. Academic Press. June 2024. doi:10.1016/B978-0-443-13244-5.00007-9.

PREPRINT

Minje Park, Carri Chan, Keith Boell, **Elliot Mitchell**, Abdul Tariq, David Vawdrey. Machine Learning-Guided Cancer Screening: The Benefits of Proactive Care. September 18, 2024. doi:10.2139/ssrn.4959547

PRESENTATIONS

CONFERENCE and WORKSHOP PRESENTATIONS

Elliot G Mitchell, Minje Park, Rebecca Maff, Abdul Tariq, Keith Boell, David Vawdrey, Carri Chan. Quantifying the Impact of Proactive ML Risk Stratification Interventions: A Regression Discontinuity Approach. American Medical Informatics Association (AMIA) Annual Symposium. November 2025. Atlanta, GA.

Alexander D. Pretko, James R. Elmore, David K. Vawdrey, Rebecca M. Maff, and **Elliot G. Mitchell**. Podium Quantifying the Impact of a Machine Learning Powered High-Risk Screening Program for Abdominal Aortic Aneurysms. *American Medical Informatics Association (AMIA) Annual Symposium.* November 2024. San Francisco, CA.

Elliot G Mitchell, Satish Kalepalli, Grant DeLong, Abdul Tariq, Vishal Mehra, Janet Tomcavage, Emily Fry, David K. Vawdrey. Iterative Development and Preliminary Evaluation of an AI-Powered Care Management Program to Reduce Unplanned Utilization. *American Medical Informatics Association*

(AMIA) Clinical Informatics Conference (CIC). ***AI Evaluation Showcase Stage II***. May 2024. Minneapolis, MN.

Biplab S. Bhattacharya, **Elliot G Mitchell**, Gaurav Shetty, Grant DeLong, Tamanna Munia, Abdul Tariq A Long COVID Risk Predictor Focused on Clinical Workflow Integration *American Medical Informatics Association (AMIA) Annual Symposium.* November 2023. New Orleans, LA.

Abdul A Tariq, **Elliot G Mitchell**, David K. Vawdrey How Much Do Social Determinants of Health, Determine Health? *American Medical Informatics Association (AMIA) Annual Symposium.* November 2023. New Orleans, LA.

Elliot G Mitchell, Satish Kalepalli, Grant DeLong, Meg Horgan, Vishal Mehra, David K. Vawdrey, Karen Murphy, Abdul Tariq. Towards Intuitive Features for Useful Explanations of Predictive Models. *American Medical Informatics Association (AMIA) Annual Symposium.* November 2022. Washington, D.C.

Mitchell EG, Mamykina L. "There For You Every Day": Pilot Study of an Automated Conversational Health Coach *American Medical Informatics Association (AMIA) Annual Symposium.* November 2021. San Diego, CA.

Chen R, Averitt A, Bear Don't Walk O, Bhave S, Grossman Liu L, **Mitchell EG**, Rodriguez V, Thangaraj P, Sun T. The CLinically Explainable Actionable Risk (CLEAR) Model. *American Medical Informatics Association (AMIA) Annual Symposium.* November 2020. (Virtual)

Mitchell EG, Mamykina L. Computational Approaches to Support Automated Conversational Health Coaching. *International Conference on AI in Medicine – Doctoral Consortium*, August, 2020. Minneapolis, MN (Virtual)

Mitchell EG, Mamykina L. Automated Conversational Health Coaching. Plenary Presentation *National Library of Medicine Informatics Training Conference*, June, 2020. Virtual.

Mitchell EG, Mamykina L. The Coach with the Most: Towards Conversational Health Coaching. *Conversational Agents for Health Workshop (CHI 2020)*, April, 2020. Virtual.

Mitchell EG, Burgermaster M, Heitkemper E, Levine M, Miao Y, Tabak E, Albers D, Smaldone A, and Mamykina L. Feasibility of a machine learning based method to generate personalized nutrition goals for diabetes self-management. *AMIA Annual Symposium*, November, 2019. Washington, DC.

Grossman LV, **Mitchell EG**, Hripcsak G, Weng C, Vawdrey DK. A method for harmonization of clinical abbreviation and acronym sense inventories. *National Library of Medicine Informatics Training Conference*, June, 2019. Indianapolis, IN. **Best Presentation Award**

Mitchell EG, Burgermaster M, Heitkemper E, Levine M, Miao Y, Desai P, Hwang M, Tabak E, Albers D, Smaldone A, and Mamykina L. 2019. Personalized, data-driven recommendations for diabetes self-management with GlucoGoalie. *Workshop on Interactive Systems in Healthcare (WISH)* May, 2019. Glasgow, UK.

Mitchell EG, Burgermaster M, Heitkemper E, Reading M, Levine ME, Miao Y, Desai P, Mamykina L. An Intelligent Voice Assistant for Diabetes Self-Management: T2D2 – Taming Type 2 Diabetes, Together. *AMIA Annual Symposium,* November, 2018. San Francisco, CA.

Mitchell EG, Burgermaster M, Heitkemper E, Mamykina L. Designing Conversational Interfaces for Diabetes Self-Management. *American Association of Diabetes Educators*, August, 2018. Baltimore, MD.

Mitchell EG, Opportunities for Conversational Voice Assistants in Chronic Disease Self-Management. *National Library of Medicine Informatics Training Conference*, June, 2018. Nashville, TN.

CONFERENCE ABSTRACTS

Facktor MA, Maff R, Keister C, **Mitchell E**, Jin Y, Ton TG, Olghi N, Obradovic M. P1. 04.31 Implementation of a Lung Cancer Risk Prediction Algorithm: Early Findings From the Lungflag™ Pilot at Geisinger Health System. Journal of Thoracic Oncology 2025 Oct 1;20(10):S120. doi:10.1016/j.jtho.2025.09.221

Facktor MA, Jin Y, Maff R, **Mitchell E**, Ton TG, Olghi N, Obradovic M. Artificial intelligence–aided lung cancer screening in routine clinical practice: A pilot of LungFlag at Geisinger. *Journal of Clinical Oncology* 42; May 2024. doi:10.1200/JCO.2024.42.16_suppl.e13604

Mitchell EG, Mamykina L. From the Curtain to Kansas: Conducting Wizard-of-Oz Studies in the Wild. *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)* May 08-13, 2021, Yokohama, Japan. Case Study. doi:10.1145/3411763.3443446

Lena Mamykina, Arlene M. Smaldone, Suzanne Bakken, Noemie Elhadad, **Elliot G. Mitchell**, Pooja M. Desai, Matthew E. Levine, Jonathan N. Tobin, Andrea Cassells, Patricia G. Davidson, David J. Albers and George Hripcsak. 2021. Scaling Up HCI Research: from Clinical Trials to Deployment in the Wild. *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts).* May 08-13, 2021, Yokohama, Japan. Case Study. doi:10.1145/3411763.3443437

Samantha Olinsky, Pooja M. Desai, Selen Turkay, Elizabeth M Heitkemper, **Elliot G Mitchell**, Lena Mamykina, and Maria L. Hwang. 2021. Meals for Monsters: a Mobile Application for the Feasibility of Gaming and Social Mechanisms. *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems.* May 08-13, 2021, Yokohama, Japan. doi:10.1145/3411763.3451789

Mitchell EG, Maimone R, Mamykina L. Characterizing Human vs. Automated Coaching: Preliminary Results. *CHI EA '20: Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems.* May 2020. doi:10.1145/3334480.3383081

Juul F, Mamykina L, Almonte A, Sepulveda J, Heitkemper E, **Mitchell EG**, Burgermaster M. User Experiences with Platano, a Dietary Self-Monitoring App for Underserved Patients with Type 2 Diabetes. *Journal of Nutrition Education and Behavior.* 2019 Jul 1;51(7):S44-5. doi:10.1016/j.jneb.2019.05.404

Burgermaster M, Juul F, **Mitchell E**, Heitkemper E, Sepulveda J, Almonte A, Mamykina L. Personal Informatics Technology for Engagement in Community Health (PI-TECH): Mixed Methods Study of Platano, a Motivationally Tailored App for Dietary Diabetes Management. *Current Developments in Nutrition*, Vol. 3, Issue Supplement_1, June 2019, nzz050. P16–051–19, doi:10.1093/cdn/nzz050.P16-051-19

POSTERS

Tamanna Tabassum Munia, **Elliot G Mitchell**, Brandy Feltman, Timothy Murphy, Robin Skrine, Rosemary Leeming, David Vawdrey, Aalpen Patel, Keith Boell, Biplab S Bhattacharya. Integrating a Breast Cancer Risk Model into a Clinical Workflow. American Medical Informatics Association (AMIA) Annual Symposium. November 2025. Atlanta, GA.

Akiva Blickstein, Casey Cauthorn, Debdipto Misra, Matthew Facktor, Rebecca Maff, **Elliot G Mitchell**, Biplab S Bhattacharya. Dashboard Development for Tracking Performance of Lung Cancer Screening. American Medical Informatics Association (AMIA) Annual Symposium. November 2025. Atlanta, GA.

Akiva Blickstein, Biplab Bhattacharya, Casey Cauthorn, Evan Ryer, Gregory Salzler, Jim Urick, Rebecca Maff, James Elmore, and **Elliot G Mitchell**. Dashboard Development for Tracking Performance of High-Risk Abdominal Aortic Aneurysm Screening. *American Medical Informatics Association (AMIA) Annual Symposium.* November 2024. San Francisco, CA.

Elliot G Mitchell, Abdul Tariq, Casey Cauthorn, David K Vawdrey, Aalpen A Patel Implementing Ad-

vanced AI in Healthcare. *New York Academy of Sciences (NYAS): The New Wave of AI in Healthcare.* May 2023. New York, NY.

Grant Delong, Henry K. Philofsky, James Elmore, Stacey Shriner, Sara Hunt, Casey Cauthorn, **Elliot G. Mitchell**, David K. Vawdrey, Abdul Tariq. SentenCy: An Open-source Tool for Improving Biomedical Named Entity Recognition. *American Medical Informatics Association (AMIA) Annual Symposium.* November 2022. Washington, D.C.

Chen R, Rodriguez V, Grossman Liu LV, **Mitchell EG**, Averitt A, Bear Don't Walk O, Bhave S, Sun T, Thangaraj P, and the Columbia DBMI CMS AI Challenge Team. Engendering Trust and Usability in Clinical Prediction of Unplanned Admissions: The CLinically Explainable Actionable Risk (CLEAR) Model. *Machine Learning for Health (MLHC).* August 2020.

Mitchell EG, Burgermaster M, Heitkemper E, Mamykina L. Perceptions of Conversational Voice Assistants Among Diabetes Care and Education Specialists. *Association of Diabetes Care and Education Specialists.* August 2020.

Mitchell EG, Mamykina L. T2Coach: A conversational agent health coach for diabetes self-management. *Data Science Day 2020: Virtual Preview.* March 2020.

Mitchell EG, Mamykina L. Towards automated conversational health coaching. *New York Academy of Sciences (NYAS) Natural Language, Dialog, and Speech (NDS) Symposium*, November 2019, New York, NY.

Mitchell EG, Levine M, Mamykina L, Tabak E, Albers D. Machine learning for personalized decision support with patient-generated health data. *AMIA Annual Symposium*, November 2019. Washington, DC.

Mitchell EG, Heitkemper E, Ensari I, Albers D, Smaldone A, Mamykina L. Data-Driven Personalization of Nutrition and Physical Activity Goals for Diabetes Self-Management. *Columbia Data Science Institute (DSI) Data Science Day*, 2019. New York, NY.

Mitchell EG, Levine ME, Miao Y, Burgermaster M, Hwang M, Heitkemper E, Smaldone A, Albers DJ, Mamykina L. Automated Generation of Personalized Nutrition Goals for Diabetes Self-Management with Platano. *Columbia Data Science Institute (DSI) Data Science Day*, 2018. New York, NY.

Mitchell EG, Levine ME, Albers DJ, Mamykina L. Reflecting on Diabetes Self-Management Logs with Simulated, Continuous Blood Glucose Curves: A Pilot Study. *AMIA Annual Symposium*, November 2017. Chicago, IL.

CHALLENGES and COMPETITIONS

Mitchell EG, Grossman LV. A Multimodal Conversational Agent to Support Diabetes Self-Management. *AMIA Annual Symposium Student Design Challenge,* 2018 **(First place)**

Grossman LV, **Mitchell EG**. Visualizing the Patient-Reported Outcomes Measurement Information System (PROMIS) Measures for Clinicians and Patients. *AMIA Annual Symposium Student Design Challenge,* 2017 **(First place)**

Mitchell EG, Burgermaster M, Heitkemper E, Reading M, Levine ME, Miao Y, Desai P, Mamykina L. Presenting T2D2: Taming Type 2 Diabetes, Together. *Alexa Diabetes Challenge Demo Day*, 2017 **(Top 5 finalist)**

Mitchell EG, Burgermaster M, Heitkemper E, Reading M, Levine ME, Miao Y, Desai P, Mamykina L. Presenting T2D2: Taming Type 2 Diabetes, Together. *HITLAB World Cup Demo Day*, 2017 **(Third place)**

INVITED TALKS

Human-Centered Design for AI in Healthcare: Applications to diabetes self-management and explainable (re)admission predictions. *Geisinger Informatics Education Seminar.* Virtual. February 2023.

Human-Centered AI for Health Coaching. *American Medical Informatics Association (AMIA) Annual Symposium.* Washington, D.C. November 2022.

Best Practices for AI Implementation in Healthcare (Discussion Group). *RE•WORK AI in Healthcare Summit*, Boston, MA. October 2022.

Deep Natural Language Processing *Geisinger Machine Learning Reading Group.* Virtual. July 2022.

From Messy Data to Useful Tools. *Columbia Data Science Institute Scholars Program.* Virtual. June 2022.

Personalizing Health Goals with Data Science. *Columbia Data Science Institute Health Analytics Center Workshop.* New York, NY. February 2020.

Voice Technology. *Society for Nutrition Education and Behavior (SNEB) Digital Tech for Nutrition Education Roundup.* Webinar. April 2019.

SERVICE and MENTORSHIP

Mentor, AMIA High School Scholars Program 2019
Attend conference with high school students pursuing informatics research

Student Planning Committee (SPC), NLM Informatics Training Conference 2019
SPC member for National Library of Medicine Informatics Training Conference in Indianapolis, IN

Assistant Editor, MedInfo2019 2019
Member of the MedInfo2019 Congress Proceedings Editorial Team

Student Representative, Columbia University DBMI, New York, NY 2018 - 2020
Elected by peers for two-year term

Co-Teaching Volunteer, TEALS, New York, NY 2018 - 2019
Introduction to Computer Science at Inwood Academy for Leadership

REVIEW ACTIVITY

American Medical Informatics Association (AMIA) Annual Symposium	2017 - Present
CHI Conference on Human Factors in Computing Systems	2018 - Present
Information Technology & People	2019 - Present
Journal of Biomedical Informatics (JBI)	2020 - Present
Journal of the American Medical Informatics Association (JAMIA)	2020 - Present
Transactions on Interactive Intelligent Systems (TiiS)	2021 - Present
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)	2022 - Present
AI and Ethics	2022 - Present

PROFESSIONAL MEMBERSHIPS

Association of Computing and Machinery, 2018 - Present
Special Interest Group on Human-Computer Interaction (SIGCHI)

American Medical Informatics Association 2016 - Present

The New York Academy of Sciences 2016 - Present

TEACHING EXPERIENCE

Teaching Assistant, Columbia University, New York, NY — Fall 2019
BINF G4008 - Intelligent Decision Support: History, Paradigms, and Applications

Workshop Organizer, The Chapin School Senior Intern Week Program — June 2019
Workshop: Evaluating the impact of interactive prototypes and systems

Co-Teaching Volunteer, TEALS, New York, NY — 2018 - 2019
Introduction to Computer Science at Inwood Academy for Leadership

Teaching Assistant, Columbia University, New York, NY — Spring 2019
BINF G6001 — Research Methods

Teaching Assistant, Columbia University, New York, NY — Spring 2018
BINF G4003 — Symbolic Methods